

AWS Academy: Cloud Architecting - Capstone Project

Lab Report

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CNG 2042 1N1: Cloud Computing

Professor [REDACTED]

2026-05-08 23:59

AWS Academy: Cloud Architecting - Capstone Project

The AWS Academy: Cloud Architecting - Capstone Project was designed to assess our ability to architect and deploy a secure, scalable, and highly-available cloud infrastructure on Amazon Web Services by applying the cloud architecture concepts and best practices covered during CNG 2042: Cloud Computing.

The Capstone Project was centered on a fictitious non-profit organization called Example Social Research Organization, that provides a website for social science researchers to obtain global development statistics, such as life expectancy and mobile phone usage data for countries around the world. The organization's website was developed by researcher Shirley Rodriguez and is built on a PHP application that queries a MySQL database containing country development statistics.

The objective of the Capstone Project was to migrate this application from a single-instance deployment into a production-ready, tiered architecture within a pre-configured Amazon Virtual Private Cloud (VPC), incorporating a managed relational database, a load-balanced application tier with automatic scaling, and secure credential management through AWS Secrets Manager. Finally, we needed to ensure that all backend systems remained inaccessible from the public internet.

Lab Processes and Procedures

Actions Taken:

Date & Time (UTC):	Source, Description, and Action Taken:
2026-05-05 21:44:49	Source: Ryan-Legion; AWS Management Console; VPC Dashboard Description: Deployed, accessed, and documented the lab environment required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: 1. From Ryan-Legion, navigated to the AWS Academy: Cloud Architecting - Capstone Project page located at https://awsacademy.instructure.com/courses/154207/assignments/1822927?module_item_id=15009401. 2. From the Capstone Project page, selected the 'Start Lab' button in the upper-right of the screen and waited for the 'AWS' button status indicator to turn green before selecting it to navigate to the AWS Management Console for the Capstone Project.

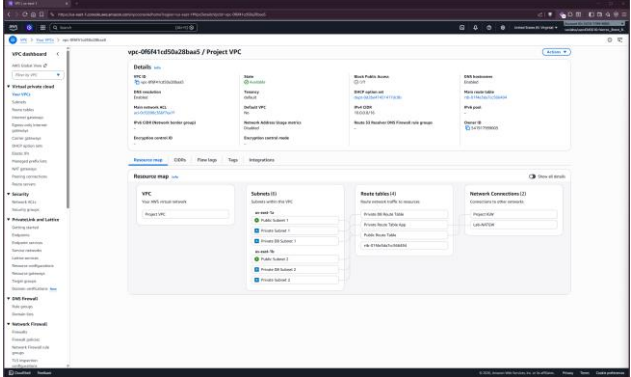
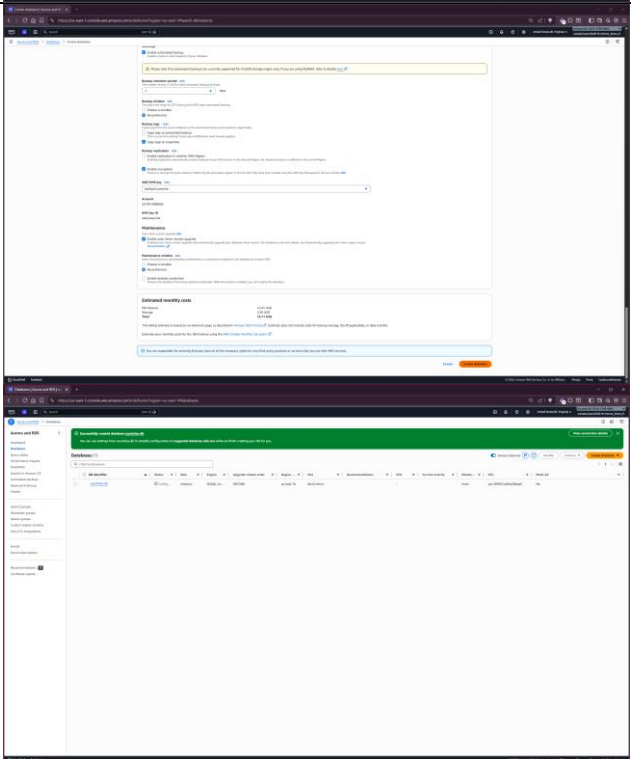
	- From the AWS Management Console, typed "VPC" in the search bar located at the top of the screen and selected the 'VPC' hyperlink to navigate to the VPC dashboard. - From the VPC dashboard, selected 'Your VPCs' from the left-navigation pane before selecting "Project VPC" on the Your VPCs page and documenting the Resource Map from the resulting screen.
2026-05-05 21:54:44	Source: Ryan-Legion; Description: Created and configured the Amazon RDS MySQL database required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: - From the AWS Management Console, typed "RDS" in the search bar located at the top of the screen and selected the 'RDS' hyperlink from the resulting search dropdown menu. - From the AWS Aurora and RDS Dashboard, selected 'Databases' from the left navigation pane to navigate to the Databases dashboard before selecting the orange 'Create Database' button from the upper-right and choosing 'Full Configuration' from the resulting dropdown. - From the Create Database page, configured an RDS Database with the following settings before selecting the 'Create Database' button located at the bottom of the screen: - Engine Type: MySQL - Templates: Dev/Test - Availability and Durability: Single-AZ DB instance deployment - DB Instance Identifier: countries-db - Master Username: admin - Credentials Management: Managed in AWS Secrets Manager - DB Instance Class: Burstable Classes - Instance Type: db.t3.micro - Allocated Storage: 20 GiB - Storage Autoscaling: Disabled - Virtual Private Cloud (VPC): Project VPC - Existing VPC Security Groups: ExampleDB-SG - Initial Database Name: countries - Enhanced Monitoring: Disabled - All other configuration settings were kept at their default values.
2026-05-05 23:18:33	Source: Ryan-Legion; Description: Created and configured the Target Group for the Application Load Balancer required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: From the AWS Management Console, typed "EC2" in the search bar located at the top of the page and selected the 'EC2'

	hyperlink from the resulting search dropdown menu to navigate to the EC2 Dashboard. - From the Amazon EC2 Dashboard, selected 'Target Groups' from the left navigation pane before clicking on the orange 'Create Target Group' option from the Target Groups page. - From the Create Target Group page, configured the Target Group with the following settings before selecting the 'Create Target Group' button: - Target Type: Instances - Target Group Name: countries-tg - VPC: Project VPC - All other configuration settings were kept at their default values.
2026-05-05 23:30:32	Source: Ryan-Legion; Description: Configured and created the Application Load Balancer required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: - From the EC2 Dashboard, selected 'Load Balancers' from the left navigation pane and clicked on the orange 'Create Load Balancer' button in the upper-right and choosing 'Application Load Balancer' from the resulting dropdown menu. - From the Create Application Load Balancer page, configured the Application Load Balancer with the following settings before selecting the 'Create Application Load Balancer' button: - Load Balancer Name: countries-alb - VPC: Project VPC - Availability Zones and Subnets: us-east-1a (Public Subnet 1); us-east-1b (Public Subnet 2) - Security Groups: ALBSG - Forward to Target Group: countries-tg - All other configuration settings were kept set to their default values, with the exception of the removal of the default security group.
2026-05-05 23:41:38	Source: Ryan-Legion; Description: Configured and created the Auto Scaling Group required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: - From the EC2 Dashboard, selected 'Auto Scaling Group' from the left navigation pane and clicked on the orange 'Create Auto Scaling Group' button from the upper-right of the page. - From the Choose Launch Template page, configured the Auto Scaling Group with the following settings before selecting the orange 'Create' button from the bottom-right of the page: - Auto Scaling Group Name: countries-asg - Launch Template: Project-LT - Version: Latest (1) - VPC: Project VPC

	e. Availability Zones and Subnets: Private Subnet 1; Private Subnet 2 f. Select Load Balancing Options: Attach to an Existing Load Balancer g. Existing Load Balancer Target Groups: countries-tg h. Additional Health Check Types: Turn on Elastic Load Balancing Health Checks i. Desired Capacity: 2 j. Minimum Desired Capacity: 2 k. Maximum Desired Capacity: 4 l. Choose Whether to Use a Target Tracking Policy: Target Tracking Scaling Policy i. All other configuration settings were kept set to their default values. 3. From the EC2 Auto Scaling Groups page, waited until the Instance Health on the countries-asg Auto Scaling Group passed 2/2 Healthy.
2026-05-06 00:13:57	Source: Ryan-Legion; Description: Connected to an instance and imported the SQL data required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: 1. From the EC2 Dashboard, selected 'Instances' from the left navigation pane before choosing the "ExampleAPP" instance running in us-east-1a with Instance ID: i-Of863a148a10b8879 and clicking the 'Connect' button in the upper-right. 2. From the resulting Connect page, selected the 'SSM Session Manager' tab before clicking on the orange 'Connect' button in the bottom-right. 3. From the resulting SSM Systems Manager terminal, utilized the command 'sudo su - ec2-user' to switch to the ec2-user account before using the command 'wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-200-ACACAD-3-113230/22-lab-Capstone-project/s3/Countrydatadump.sql' to download the required Countrydatadump.sql file. 4. From the SSM Systems Manager terminal, utilized the command 'aws secretsmanager list-secrets --query "SecretList[*].Name" --output text' to identify the name of the secret as rds!db-b987f854-6f9f-42a5-85f9-fd5b9274024c. 5. From the SSM Systems Manager terminal, utilized the command 'aws secretsmanager get-secret-value --secret-id 'rds!db-b987f854-6f9f-42a5-85f9-fd5b9274024c' --query SecretString --output text' to identify the username and password in JSON output as {"username":"admin","password":"dwrN)8Hln)Lrzv4Ns07HbSdiT:~Q"}.

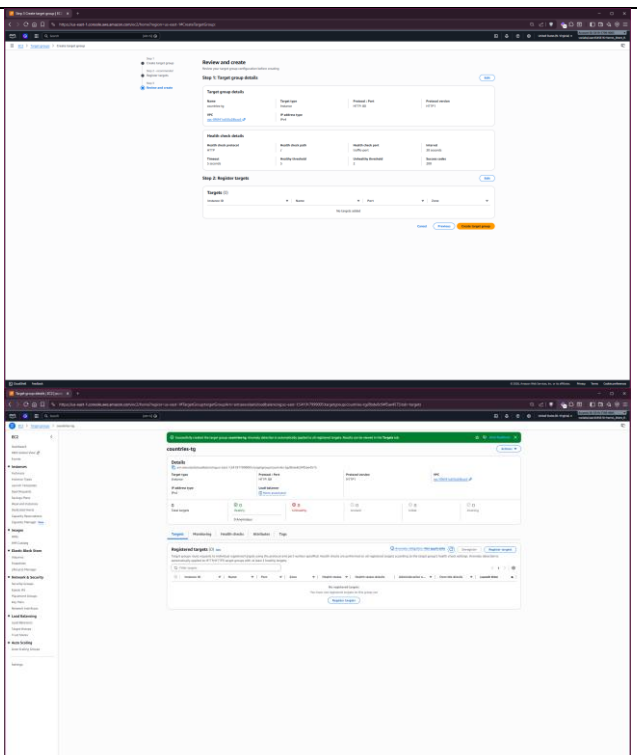
	- From the SSM Systems Manager terminal, utilized the command 'mysql -h countries-db.cjssw6k0pu7e.us-east-1.rds.amazonaws.com -u admin -p countries < /home/ec2-user/Countrydatadump.sql' and input the previously identified admin password to import the data from the Countrydatadump.sql file to the countries-db database. - From the SSM Systems Manager terminal, utilized the command 'mysql -h countries-db.cjssw6k0pu7e.us-east-1.rds.amazonaws.com -u admin -p countries' and entered the password dwrN)8Hln)Lrzv4Ns07HbSdiT:~Q to access the mysql database. - From the MySQL command-line, utilized the command 'SHOW TABLES;' to list the tables where I was able to observe the countrydata_final table. - From the MySQL command-line, utilized the command 'SELECT * FROM countrydata_final LIMIT 5;' for additional confirmation that data was present on database before utilizing the 'exit' command to leave the MySQL command-line terminal.
2026-05-06 01:06:29	Source: Ryan-Legion; Description: Tested the Application Load Balancer URL and Web Application Query page to ensure functionality required to complete the AWS Academy: Cloud Architecting - Capstone Project. Action Taken: - From the EC2 Dashboard, selected 'Load Balancers' from the left navigation pane before choosing the countries-alb load balancer where I was able to observe that the DNS name was countries-alb-269316742.us-east-1.elb.amazonaws.com. - From a new browser tab, navigated to the web application located at http://countries-alb-269316742.us-east-1.elb.amazonaws.com/index.php and clicked on the 'Query' hyperlink in the upper-left of the page. - From the resulting Query page, selected 'Mobile Phones' and 'Population' from the dropdown menu, clicking on submit each time to test functionality of the application, which resulted in me being able to observe output from each query.
2026-05-06 01:47:34	Source: Ryan Legion; Description: Submitted the AWS Academy: Cloud Architecting - Capstone Project for grading. Action Taken: From the Capstone Project page, selected the 'Submit' button from the upper-right of the screen. The grade returned as a 35/35 (100%), bringing the AWS Academy: Cloud Architecting - Capstone Project to a conclusion. Additionally, the Grades page in the AWS Academy courseware split the Capstone Project into two entries with one indicating 35/35 and another indicating 56/56.

Evidence Log:

Date & Time (UTC):	Action Taken:	Evidence:
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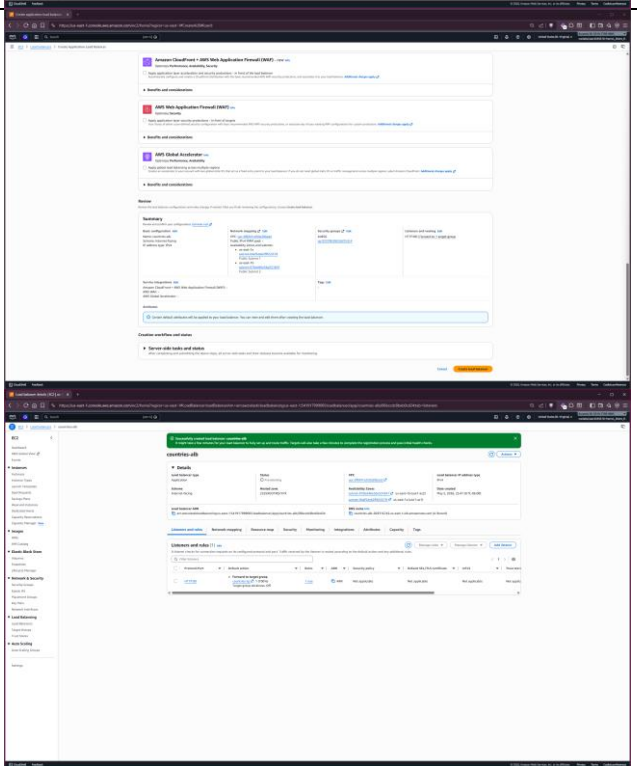
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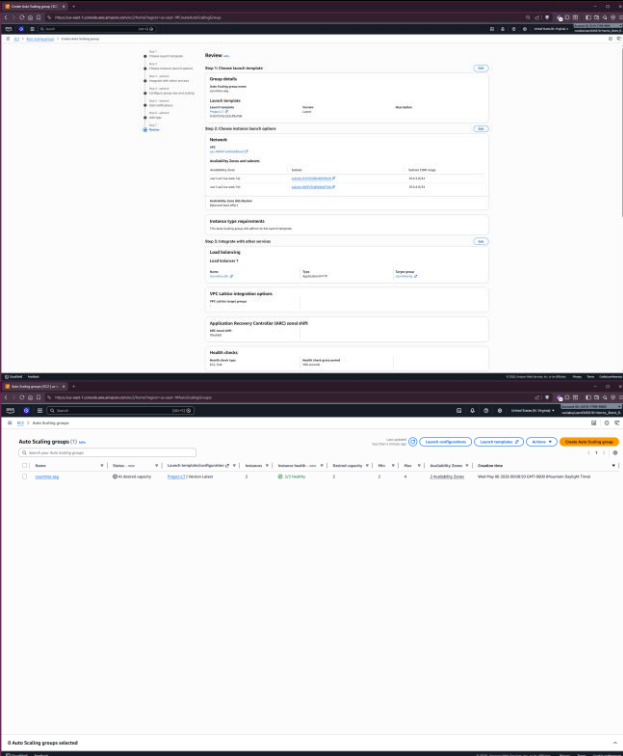
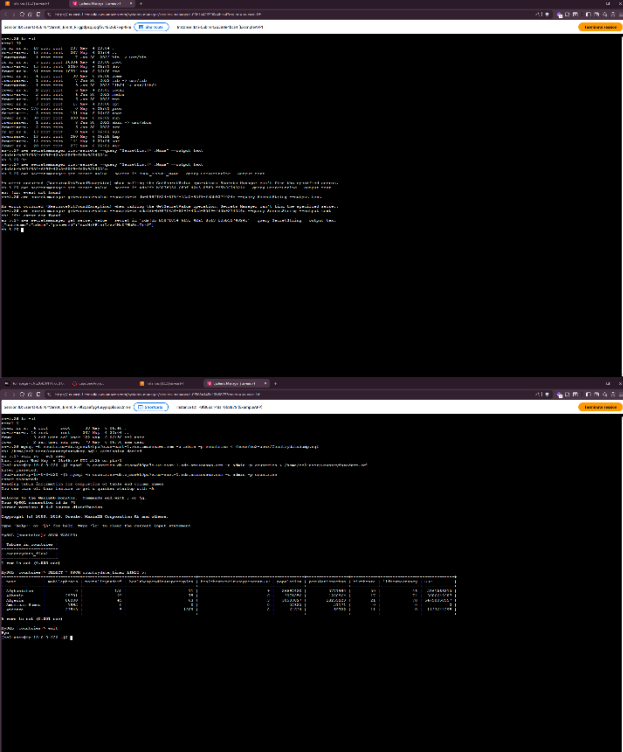
Created and configured the Target Group for the Application Load Balancer required to complete the AWS Academy: Cloud Architecting - Capstone Project.



2026-05-05 23:30:32

Configured and created the Application Load Balancer required to complete the AWS Academy: Cloud Architecting - Capstone Project.



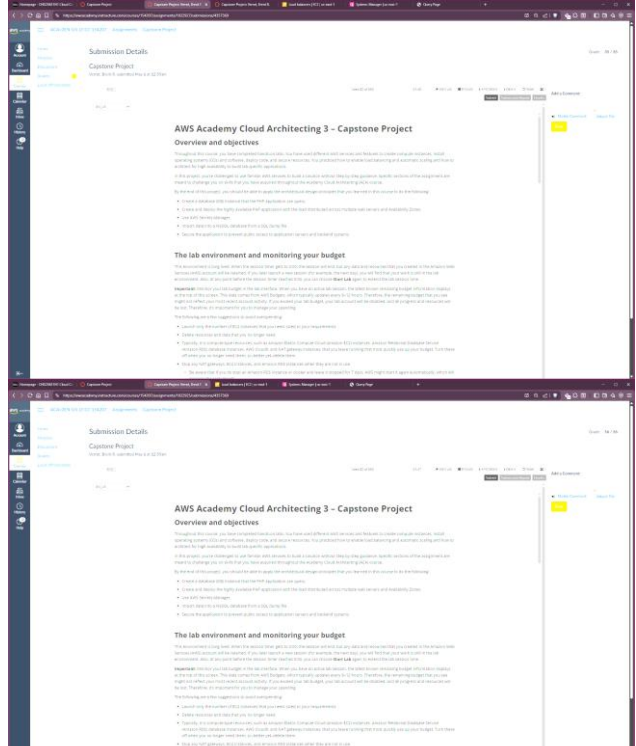
2026-05-05 23:41:38	Configured and created the Auto Scaling Group required to complete the AWS Academy: Cloud Architecting - Capstone Project.	
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2026-05-06 01:06:29

Tested the Application Load Balancer URL and Web Application Query page to ensure functionality required to complete the AWS Academy: Cloud Architecting - Capstone Project.

The screenshot displays a web application interface for an 'Example Social Research Organization'. The top section, titled 'About Us', features a profile picture of a woman and a brief description. Below this is a 'Contact Us' section with a form and a 'Home' button. The main content area is the 'Country Data Query Page', which includes a prompt to 'Please select which query you want to run.' and a dropdown menu. The bottom section shows a table of data for various countries, with columns for 'Country Name', 'Population', and 'GDP (USD)'. The table lists countries such as Canada, Mexico, USA, Brazil, India, China, and others, along with their respective population and GDP values.

Country Name	Population	GDP (USD)
Canada	38,210,761	1,559,927,000,000
Mexico	128,132,963	1,210,000,000,000
USA	331,002,651	21,427,000,000,000
Brazil	215,012,321	1,736,000,000,000
India	1,380,004,385	2,874,000,000,000
China	1,402,147,375	14,185,000,000,000
Argentina	45,597,575	448,000,000,000
Japan	126,478,979	5,051,000,000,000
South Korea	51,269,185	1,639,000,000,000
France	67,390,136	2,775,000,000,000
Germany	83,000,000	3,846,000,000,000
UK	67,334,743	2,775,000,000,000
Italy	60,271,824	2,145,000,000,000
Spain	46,754,776	1,414,000,000,000
Sweden	10,089,339	514,000,000,000
Poland	38,082,000	1,140,000,000,000
Belgium	11,589,425	549,000,000,000
Portugal	10,642,349	254,000,000,000
Canada (GDP)	1,559,927,000,000	1,559,927,000,000
Mexico (GDP)	1,210,000,000,000	1,210,000,000,000
USA (GDP)	21,427,000,000,000	21,427,000,000,000
Brazil (GDP)	1,736,000,000,000	1,736,000,000,000
India (GDP)	2,874,000,000,000	2,874,000,000,000
China (GDP)	14,185,000,000,000	14,185,000,000,000
Argentina (GDP)	448,000,000,000	448,000,000,000
Japan (GDP)	5,051,000,000,000	5,051,000,000,000
South Korea (GDP)	1,639,000,000,000	1,639,000,000,000
France (GDP)	2,775,000,000,000	2,775,000,000,000
Germany (GDP)	3,846,000,000,000	3,846,000,000,000
UK (GDP)	2,775,000,000,000	2,775,000,000,000
Italy (GDP)	2,145,000,000,000	2,145,000,000,000
Spain (GDP)	1,414,000,000,000	1,414,000,000,000
Sweden (GDP)	514,000,000,000	514,000,000,000
Poland (GDP)	1,140,000,000,000	1,140,000,000,000
Belgium (GDP)	549,000,000,000	549,000,000,000
Portugal (GDP)	254,000,000,000	254,000,000,000

2026-05-06 01:47:34	Submitted the AWS Academy: Cloud Architecting - Capstone Project for grading.	
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Troubleshooting and Issues

I encountered no significant issues while completing the AWS Academy: Cloud Architecting - Capstone Project. The recommended references provided at each step of the Capstone Project, combined with the experience gained from the previous Guided Labs and Challenge Labs completed throughout the course, proved sufficient to complete the project with minimal difficulties. The only issue I encountered was the lab environment session expiring while I was working on wrapping up the Capstone Project, which was easily rectified by selecting the 'Start Lab' button and waiting for the environment and my created resources to redeploy.

Lessons Learned

The AWS Academy: Cloud Architecting - Capstone Project provided me with the opportunity to apply core cloud architecture concepts and best practices covered throughout CNG 2042: Cloud Computing in practical, simulated production environment.

The project reinforced the importance of proper prioritization when deploying a multi-tier architecture since each component had dependencies on the resources created before it. The experience gained from the previous Guided Labs and Challenge Labs through the course, especially the labs covering highly available environments and Amazon RDS database configuration, provided me with the tools that made the configuration and deployment process straightforward and easy to accomplish. Additionally, the Capstone Project highlighted the benefit of AWS Secrets Manager as a secure alternative to hardcoded credentials and demonstrated how managed credential storage can be integrated directly into database provisioning and retrieved from the CLI while avoiding exposure of sensitive information.

Finally, the successful completion of the AWS Academy: Cloud Architecting - Capstone Project with a grade of 35/35 (100%) demonstrated proficiency in designing and deploying a secure, scalable, and highly-available cloud architecture on Amazon Web Services.